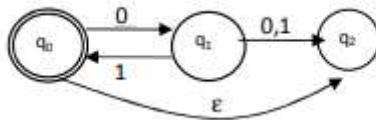
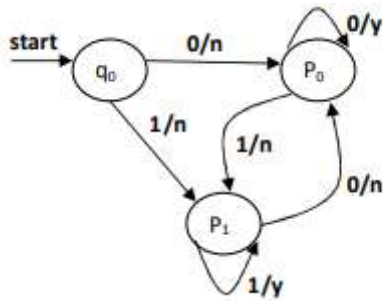


Unit I

1. Explain the design of a finite state machine with an example?
2. Explain the advantages of Finite State Machine?
3. What is NFA? Explain the transitions of NFA?
4. Design DFA to accept strings with 'c' and 'd' such that number of 'd' should be at end of the string.
5. Show with an example equivalence between NFA with and without ϵ -transitions
6. Construct an NFA that accepts the set of all strings over $\{0,1\}$ that start with 0 or 1 and end with 10 or 01.
7. Let $\Sigma = \{a, b\}$,
 - a) Give DFA that accepts any string with aababb as a substring.
 - b) Minimize the DFA obtained for the answer of question (a).
8. $(0/1)^*011$ for this regular expression draw the NFA with ϵ -closures and convert it into NFA.
9. Construct a DFA equivalent to the NFA given below



10. Convert the following Mealy machine to an equivalent Moore machine



11. Differentiate between NFA and DFA?
- 12.

Construct Minimum state Automata for the following DFA?

* denotes final state

δ	0	1
$\rightarrow q1$	q2	q6
q2	q1	q3
*q3	q2	q4
q4	q4	q2
q5	q4	q5
*q6	q5	q4

Unit II

1. Construct a DFA for the Regular Language consisting of any number of a's and b's
2. What is a regular language? Convert the given regular expression to regular language.

- i) $(1+\epsilon)(00^*1)0^*$
- ii) $(0^*1^*)000(0+1)^*$
- iii) $(00+10)^*1^*(10+00)^*$

3. What is relationship between finite automata and regular expression? Explain the process of converting DFA to regular expression.
4. Let $\Sigma = \{a, b\}$,
 - a) Give DFA that accepts any string with aababb as a substring.
 - b) Minimize the DFA obtained for the answer of question 2(a) using MyhillNerode theorem.
5. Construct DFA equivalent to regular expression $(0+1)^*(00+11)(0+1)^*$ and also find the reduced DFA
6. Explain the Pumping Lemma for regular sets. Show that $L=\{a^p \mid p \text{ is a prime}\}$ is not regular.